SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

RETROX

- Team Name : Retro_Link
- Theme : Open Innovation
- PS Category : Software & Hardware
- Team ID : DEX26161

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RETROX – RETROfit eXcellence

PROBLEM STATEMENT :

Manual machines remain useful for one-off parts, prototypes, repairs and quick stock preparation. Yet their status, cycle counts, downtime and condition often depend on operator observation and manual records.

WHY MANUAL MACHINES STILL MATTER

- No CNC programming needed for many quick jobs
- Flexible for low-volume work and repairs
- Useful for secondary operations and training

THE MONITORING GAP

- Limited live visibility into machine usage
- Manual cycle and downtime tracking
- Less historical data for supervision and maintenance

THE RETROX CONCEPT :

A low-cost, non-invasive retrofit kit for small and mid-scale industries. External sensors and an ESP32 collect selected machine data and send it to a Wi-Fi dashboard—while the original controls and manual operation remain unchanged.

HOW IT WORKS :

MANUAL MACHINE — Existing equipment

SENSORS — Capture signals

ESP32 — Process data

DASHBOARD — Status & alerts

Target: conventional, manually operated machines—not autonomous CNC machines.

Technical Approach

SENSORS & MACHINE SIGNALS

Current • vibration • contactless IR temperature
Magnetic proximity for spindle RPM • Hall effect
Laser + LDR beam interruption in running zone
Float switch for coolant / water level

ESP32 CONTROLLER

Reads inputs • validates sensor data
Calculates RPM, machine status and metrics
Runs local logic and triggers alerts

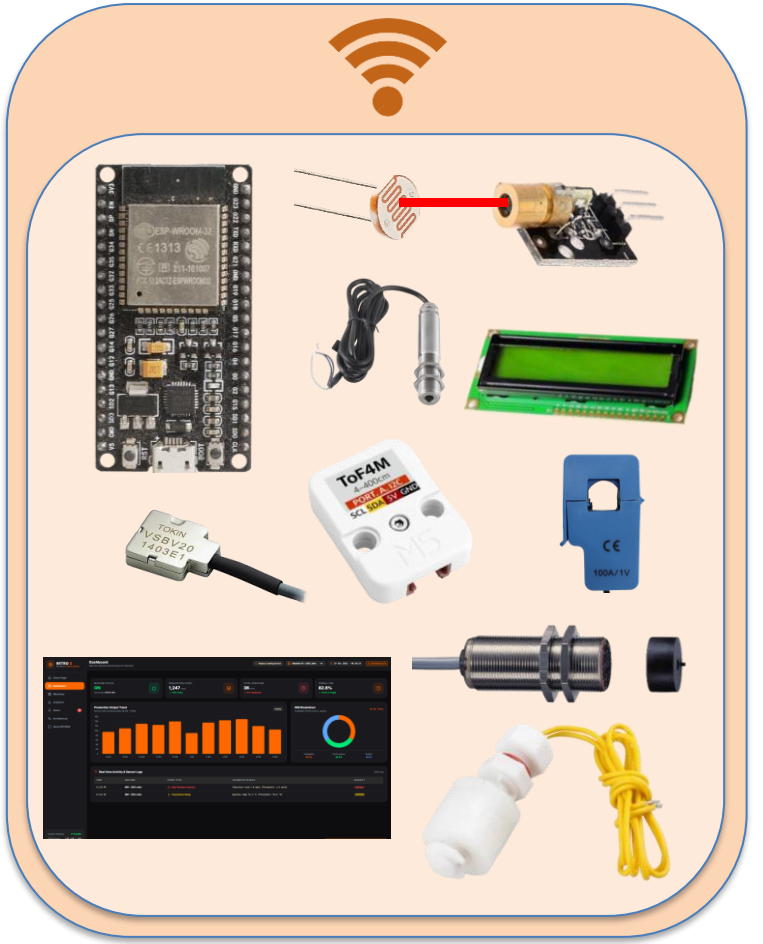
SAFE INTERFACING

Modular device, designed not to interfere with machine operation. Beam interruption raises an alert; use a safety-rated system for personnel protection

WI-FI + DASHBOARD

ESP32 Wi-Fi transmits readings to the IoT dashboard. Displays status, trends, alerts and downtime analysis

LOCAL DISPLAY 16×2 LCD • live RPM, temperature, vibration, current, machine status and liquid-level alerts



Software stack

- Embedded C/C++ for sensor acquisition and local logic
- HTML, CSS and JavaScript for the dashboard
- Threshold-based abnormality alerts; OEE calculated only when required data is available

Feasibility & Viability

ESP32

TECHNICAL FEASIBILITY

Uses affordable ESP32 hardware and readily available sensors.

Supports external monitoring without replacing the original controller.

Adaptable to different machines through calibration.

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CHALLENGES AND RISKS

Sensor compatibility and inaccurate readings.

Electrical noise and Wi-Fi interruptions.

Safe installation on industrial equipment.

✓

MITIGATION STRATEGIES

Use isolated interfaces and calibrated sensors.

Apply signal filtering and data validation.

Add local buffering and a protective enclosure.

VIABILITY

Low-cost and modular retrofit kit.

Scalable monitoring solution for small and mid-scale industries.

Business Canvas

MISSION & VISION

Make legacy machines smarter through affordable retrofitting and bring Industry 4.0 to MSMEs.

MARKET & COMPETITION

Small and medium-sized manufacturers, with an affordable alternative to costly industrial IoT systems.

BUSINESS MODEL



Various Models for different machines. like with Lubrication Level Monitoring and without it, installation services, and optional dashboard subscriptions.

FINANCIAL PLANNING

Costs: hardware and development.
Revenue: kits, installation and services.

Mission: make legacy machines smarter through affordable, retrofit-first monitoring.

Impact & Benefits

RETROX helps small and mid-scale industries improve visibility into conventional manual machines — without replacing the original controller.

ECONOMIC

Lower modernization costs
Potentially reduce production losses

Value: retrofit monitoring instead of a full machine upgrade.

OPERATIONAL

Better visibility of cycles, downtime and machine utilization

Value: more informed supervision and maintenance decisions.

SOCIAL

Supports digital skills and technology adoption in MSMEs

Value: helps teams build familiarity with machine data.

ENVIRONMENTAL

May extend useful machine life and reduce premature disposal

Value: retrofit existing equipment rather than replace it.

OVERALL IMPACT

Makes practical Industry 4.0 monitoring more accessible to MSMEs

Scope: monitoring support, not autonomous machine control.

RETROX turns existing machine data into useful visibility—supporting better supervision, not replacing skilled operators.

Research & References

ESP32 Documentation
Technical specifications, libraries and development guides.
https://docs.espressif.com/

Modbus Industrial Communication
Protocol details and implementation in industrial systems.
https://www.modbus.org/

OEE Calculation
Methodology and best practices for manufacturing efficiency.
https://www.oee.com/

Vibration Monitoring
Technical articles and sensor design for vibration analysis.
https://www.analog.com/en/resources/technical-articles/high-fidelity-vibration-acquisition-platform.html

Energy Monitoring
Open-source hardware and documentation for energy measurement.
https://docs.openenergymonitor.org/

References support sensor integration, machine monitoring and OEE methodology.

GITHUB REPOSITORY

Repository includes circuit diagrams, web dashboard, and documentation.

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https://github.com/Anirudh-sha/RETROX-Industrial-Iot

WEB DASHBOARD

https://anirudh-sha.github.io/RETROX-Industrial-Iot/